**PROGRAMME STRUCTURE FOR BACHELOR OF SCIENCE IN DECISION MATHEMATICS AND DATA ANALYTICS WITH HONOURS [BSc. Decision Mathematics and Data Analytics (Hons)]**

The Bachelor of Science in Decision Mathematics and Data Analytics with Honours consists of the following **SIX (6)** components:

| COMPONENT | CREDIT HOURS |
|---|---|
| A. UNIVERSITY CORE COURSES | 12 |
| B. ENGLISH CORE COURSES | 6 |
| C. PROGRAMME CORE COURSES | 70 |
| D. PROGRAMME ELECTIVES | 12 |
| E. FIELD ELECTIVES | 18 |
| F. FREE ELECTIVE | 3 |
| **TOTAL CREDIT HOURS** | **121 HOURS** |

To be awarded Bachelor of Science in Decision Mathematics and Data Analytics with Honours, a student is required to take and pass a minimum of 121 credit hours which include courses in the programme structure through the following structure:

## A. UNIVERSITY CORE COURSES (12 CREDIT HOURS)

i. Malaysian Students

| CODE | COURSE | CREDIT HOURS |
|---|---|---|
| MPUK3112 | Falsafah dan Cabaran Semasa | 2 |
| MPUK3122 | Malaysian Nation-State Aspiration | 2 |
| MPUK3312 | Nationhood Studies | 2 |
| MPUK3212 | Principles of Entrepreneurship | 2 |
| MPUK3152 | Integrity And Anti- Corruption | 2 |
| MPUK3411 | Bahasa Melayu Ilmiah<br>**audit course* | 1 |
| VXXXXXXX | Co-Curriculum | 2 |
| | **TOTAL** | **13** |

ii. International Students

| CODE | COURSE | CREDIT HOURS |
|---|---|---|
| MPUK3142 | Malaysian CITRA | 2 |
| MPUK3132 | Basic Malay for Foreigner | 2 |
| MPUK3312 | Nationhood Studies | 2 |
| MPUK3212 | Principles of Entrepreneurship | 2 |
| MPUK3152 | Integrity And Anti- Corruption | 2 |
| VXXXXXXX | Co-Curriculum | 2 |
| | **TOTAL** | **12** |

- Malaysian students are **REQUIRED** to take **MPUK3411 Bahasa Melayu Ilmiah** as a compulsory audit course and will not calculated in CGPA.
- All students must choose ONE (1) co-curricular area of interest, and fulfill the requirements of that chosen area.

## B. ENGLISH CORE COURSES (6 CREDIT HOURS)

Students are required to take the English Core Courses based on their MUET/IELTS/TOEFL results obtained during the admission application. Refer to **APPENDIX A.**

## C. PROGRAMME CORE COURSES (70 CREDIT HOURS)

| Code | Course | Credit Hours | Pre-requisite |
|---|---|---|---|
| BPMNK1013 | Principles of Management | 3 | NONE |
| BKANK1013 | Basic Accounting | 3 | NONE |
| SQITK1013 | Programming in Business Applications | 3 | NONE |
| SQITK3043 | Data Modeling and Database | 3 | SQITK1013 Programming in Business Applications |
| SQITK3053 | Computer Aided Decision Making | 3 | SQITK1013 Programming in Business Applications |
| SQITK3063 | Data Mining | 3 | NONE |
| SQQPK1023 | Linear and Non-Linear Optimization | 3 | NONE |
| SQQPK2023 | Deterministic and Probabilistics Modelling | 3 | SQQPK1023 Linear and Non-Linear Optimization |
| SQQPK3033 | Computer Modeling in Business | 3 | NONE |
| SQQPK3043 | Heuristic Techniques | 3 | SQQPK1023 Linear and Non-Linear Optimization |
| SQQPK3063 | System Dynamics Modeling | 3 | NONE |
| SQQPK3083 | Multicriteria Decision Analysis | 3 | SQQPK1023 Linear and Non-Linear Optimization |
| SQQPK3103 | Operations Engineering | 3 | SQQPK1023 Linear and Non-Linear Optimization |
| SQQPK3123 | Research Methodology in Quantitative Sciences | 3 | SQITK3063 Data Mining |
| SQQZK4993 | Academic Project | 3 | SQQPK3123 Research Methodology in Quantitative Sciences AND have taken and passed AT LEAST 95 credit hours |
| SQQMK1034 | Calculus I | 4 | NONE |
| SQQMK1063 | Discrete Mathematics | 3 | NONE |
| SQQMK2023 | Linear Algebra | 3 | SQQMK1034 Calculus I OR SQQMK1023 Managerial Mathematics |
| SQQMK2033 | Intermediate Calculus | 3 | SQQMK1034 Calculus I |
| SQQSK1043 | Probability and Statistics | 3 | SQQMK1034 Calculus I |
| SQQSK2063 | Inferential Statistics | 3 | SQQSK1043 Probability and Statistics |
| SQQXK4946 | Industrial Training | 6 | Have taken ALL courses under the programme structure |

## D. PROGRAMME ELECTIVES (12 CREDIT HOURS)

Students must choose **FOUR (4)** courses from the list below:

| Code | Course | Credit Hours | Pre-requisite |
|---|---|---|---|
| SQITK3073 | Business Analytics Programming | 3 | SQITK1013 Programming in Business Applications |
| SQITK3083 | Data Visualization | 3 | NONE |
| SQQPK3113 | Game Theory and Strategic Thinking | 3 | SQQMK1034 Calculus I |
| SQQPK4013 | Productivity and Efficiency Analysis | 3 | SQQPK2023 Deterministic and Probabilistics Modelling AND SQQPK3083 Multicriteria Decision Analysis |
| SQQMK1073 | Business Mathematics | 3 | SQQMK1034 Calculus I |
| SQQSK2073 | Regression Modelling | 3 | SQQSK2063 Inferential Statistics |
| SQQSK2083 | Sampling Techniques | 3 | SQQSK2063 Inferential Statistics |

## E. FIELD ELECTIVES (18 CREDIT HOURS)

The programme offers **SIX (6)** field electives. Students are required to choose **ONE (1)** field and complete **SIX (6)** courses within the chosen field to fulfil the programme structure requirement for the graduation.

### 1. Innovative Multimedia

Choose **SIX (6)** courses from the list below:

| Code | Course | Credit Hours | Pre-requisite |
|---|---|---|---|
| SCCVK1023 | Multimedia System Foundation | 3 | NONE |
| SCCVK1043 | Visual Design Principle | 3 | NONE |
| SCCVK1053 | Perception Studies | 3 | NONE |
| SCCVK2013 | Human Computer Interaction | 3 | SCCVK1023 Multimedia System Foundation |
| SCCVK2023 | Graphic and Computer Animation | 3 | SCCVK1023 Multimedia System Foundation |
| SCCVK3043 | Technology Video Digital | 3 | NONE |

### 2. Contemporary Marketing

Choose **SIX (6)** courses from the list below:

| Code | Course | Credit Hours | Pre-requisite |
|---|---|---|---|
| BPMMK1013 | Principles of Marketing | 3 | NONE |
| BPMMK2023 | Marketing Management | 3 | BPMMK1013 Principles of Marketing |
| BPMMK3023 | Sales Management | 3 | BPMMK1013 Principles of Marketing |
| BPMMK3083 | Service Marketing | 3 | BPMMK1013 Principles of Marketing |
| BPMMK3313 | Consumer Behavior and Analytic | 3 | BPMMK1013 Principles of Marketing |
| BPMMK3333 | Global Marketing | 3 | BPMMK1013 Principles of Marketing |
| BPMMK3353 | Digital Marketing | 3 | BPMMK1013 Principles of Marketing |

### 3. Human Resource Management

Choose **SIX (6)** courses from the list below:

| Code | Course | Credit Hours | Pre-requisite |
|---|---|---|---|
| BPMHK2013 | Human Resource Management | 3 | BPMNK1013 Principles of Management |
| BPMHK3023 | Staffing | 3 | BPMHK2013 Human Resource Management |
| BPMHK3033 | Training and Development | 3 | BPMHK2013 Human Resource Management |
| BPMHK3053 | Employee Performance Management | 3 | BPMHK2013 Human Resource Management |
| BPMHK3063 | Change Management | 3 | BPMHK2013 Human Resource Management |
| BPMHK3083 | Introduction to Occupational Safety and Health | 3 | BPMHK2013 Human Resource Management |
| BPMHK3113 | International Human Resource Management | 3 | BPMHK2013 Human Resource Management |

### 4. Logistics and Transportation

Choose **SIX (6)** courses from the list below:

| Code | Course | Credit Hours | Pre-requisite |
|---|---|---|---|
| BJTLK2013 | Fundamental of Transport and Logistics | 3 | NONE |
| BJTLK3013 | Transport Planning & Policy | 3 | BJTLK2013 Fundamental of Transport and Logistics |
| BJTLK3053 | Road Transports Management | 3 | BJTLK2013 Fundamental of Transport and Logistics |
| BJTLK3113 | Logistics Management | 3 | NONE |
| BJTLK3083 | Maritime Management | 3 | BJTLK2013 Fundamental of Transport and Logistics |
| BJTLK3093 | Air Transport Management | 3 | BJTLK2013 Fundamental of Transport and Logistics |

**5. Finance**

Choose **SIX (6)** courses from the list below:

| Code | Course | Credit Hours | Pre-requisite |
|---|---|---|---|
| BWFFK2033 | Financial Management | 3 | BKALK1013 Business Accounting<br>OR<br>BKAFK1023 Introduction to Financial Accounting<br>OR<br>BKARK1013 Financial Accounting and Reporting I<br>OR<br>BKANK1013 Basic Accounting |
| BWFFK2043 | Advance Financial Management | 3 | BWFFK2033 Financial Management |
| BWFFK2053 | Financial Statement Analysis and Valuation | 3 | BWFFK2043 Advance Financial Management |
| BWFFK3013 | Corporate Finance | 3 | BWFFK2043 Advance Financial Management |
| BWFFK3033 | Financial Market and Institutions | 3 | BWFFK2033 Financial Management |
| BWFFK3073 | Personal Finance | 3 | BWFFK2043 Advance Financial Management |
| BWFNK3013 | Investment Analysis | 3 | BWFFK2043 Advance Financial Management |

**6. Bank Management**

Choose **SIX (6)** courses from the list below:

| Code | Course | Credit Hours | Pre-requisite |
|---|---|---|---|
| BWBBK1013 | Foundations of Banking | 3 | NONE |
| BWBBK2013 | Bank Management | 3 | NONE |
| BWBBK3013 | Commercial Bank Operations | 3 | BWBBK2013 Bank Management |
| BWBBK3023 | Lending Management | 3 | BWBBK2013 Bank Management |
| BWBBK3043 | Banking Securities | 3 | BWBBK2013 Bank Management |
| BWBBK3063 | International Trade and Finance | 3 | BWBBK2013 Bank Management |
| BWBBK3083 | Corporate Banking | 3 | BWBBK2013 Bank Management |

**F. FREE ELECTIVE (3 CREDIT HOURS)**

Students can choose **ONE (1)** course offered by **OTHER SCHOOLS** subject to the following conditions;

i. Must meet the prerequisite of the courses if such courses require prerequisites.
ii. **University Core, English Core** and **Foreign Language courses CANNOT** be considered as free elective courses.
iii. Equivalent courses **CANNOT** be considered as free elective courses. Refer **APPENDIX B.**

**PROGRAMME SUGGESTED COURSES BY SEMESTER FOR**
**BACHELOR OF SCIENCE IN DECISION MATHEMATICS AND DATA ANALYTICS WITH HONOURS**
**[BSc. Decision Mathematics and Data Analytics (Hons)]**

| Code | 1st Semester | Credit Hours | Code | 2nd Semester | Credit Hours |
|---|---|---|---|---|---|
| MPBXXXX/ SBLEKXXXX | English Core | 3 | MPBXXXX/ SBLEKXXXX | English Core | 3 |
| SQQPK1023 | Linear and Non-Linear Optimization | 3 | SQQMK2033 | Intermediate Calculus | 3 |
| SQQMK1034 | Calculus I | 4 | SQQSK1043 | Probability and Statistics | 3 |
| SQQMK1063 | Discrete Mathematics | 3 | SQITK1013 | Programming in Business Applications | 3 |
| MPUK3212 | Principles of Entrepreneurship | 2 | BKANK1013 | Basic Accounting | 3 |
| BPMNK1013 | Principles of Management | 3 | MPUK3122 / **MPUK3132 | Malaysian Nation-State Aspiration/ **Basic Malay for Foreigner | 2 |
| VXXXXXXX | Co-Curriculum I | 1 | MPUK3152 | Integrity and Anti-Corruption | 2 |
| MPUK3411 | Bahasa Melayu Ilmiah | 1 | VXXXXXXX | Co-Curriculum II | 1 |
| **TOTAL** | | **20** | **TOTAL** | | **20** |
| **Code** | **3rd Semester** | **Credit Hours** | **Code** | **4th Semester** | **Credit Hours** |
| MPUK3112 / **MPUK3142 | Falsafah dan Cabaran Semasa / **Malaysian CITRA | 2 | MPUK1042 | Nationhood Studies | 2 |
| SQQPK2023 | Deterministic and Probabilistics Modelling | 3 | SQITK3053 | Computer Aided Decision Making | 3 |
| SQQSK2063 | Inferential Statistics | 3 | SQITK3063 | Data Mining | 3 |
| SQITK3043 | Data Modeling and Database | 3 | SQQMK2023 | Linear Algebra | 3 |
| SQQPK3063 | System Dynamics Modeling | 3 | SQQPK3083 | Multicriteria Decision Analysis | 3 |
| XXXXXXXXX | Programme Elective I | 3 | XXXXXXXXX | Programme Elective II | 3 |
| XXXXXXXXX | Field Elective I | 3 | XXXXXXXXX | Field Elective II | 3 |
| **TOTAL** | | **20** | **TOTAL** | | **20** |
| **Code** | **5th Semester** | **Credit Hours** | **Code** | **6th Semester** | **Credit Hours** |
| SQQPK3033 | Computer Modeling in Business | 3 | SQQZK4993 | Academic Project | 3 |
| SQQPK3123 | Research Methodology in Quantitative Sciences | 3 | XXXXXXXXX | Programme Elective III | 3 |
| SQQPK3043 | Heuristic Techniques | 3 | XXXXXXXXX | Programme Elective IV | 3 |
| SQQPK3103 | Operations Engineering | 3 | XXXXXXXXX | Field Elective V | 3 |
| XXXXXXXXX | Field Elective III | 3 | XXXXXXXXX | Field Elective VI | 3 |
| XXXXXXXXX | Field Elective IV | 3 | XXXXXXXXX | Free Elective | 3 |
| **TOTAL** | | **18** | | **TOTAL** | **18** |
| **Code** | **6th Semester (Short Semester)** | **Credit Hours** | | | |
| SQQXK4946 | Industrial Training | 6 | | | |
| **TOTAL** | | **6** | | | |

**Total Overall Credit Hours: 121 Credit Hours**

**Compulsory for International Student.

**PROGRAMME STRUCTURE FOR BACHELOR OF SCIENCE WITH HONOURS (BUSINESS MATHEMATICS)**
**[B.Sc. (Hons) (Business Mathematics)]**

The Bachelor of Science with Honours (Business Mathematics) consists of the following **SIX (6)** components:

| COMPONENT | CREDIT HOURS |
|---|---|
| A. UNIVERSITY CORE COURSES | 12 |
| B. ENGLISH CORE COURSES | 6 |
| C. PROGRAMME CORE COURSES | 69 |
| D. BUSINESS PROGRAMME ELECTIVE | 15 |
| E. QUANTITATIVE PROGRAMME ELECTIVE | 15 |
| F. FREE ELECTIVE | 3 |
| **TOTAL CREDIT HOURS** | **120 HOURS** |

To be awarded the Bachelor of Science with Honours (Business Mathematics), a student is required to take and pass a minimum of **120 credit hours** which include courses in the programme structure through the following structure:

## A. UNIVERSITY CORE COURSES (12 CREDIT HOURS)

i. Malaysian Students

| CODE | COURSE | CREDIT HOURS |
|---|---|---|
| MPUK3112 | Falsafah dan Cabaran Semasa | 2 |
| MPUK3122 | Malaysian Nation-State Aspiration | 2 |
| MPUK3312 | Nationhood Studies | 2 |
| MPUK3212 | Principles of Entrepreneurship | 2 |
| MPUK3152 | Integrity And Anti- Corruption | 2 |
| MPUK3411 | Bahasa Melayu Ilmiah<br>**audit course* | 1 |
| VXXXXXXX | Co-Curriculum | 2 |
| | **TOTAL** | **13** |

ii. International Students

| CODE | COURSE | CREDIT HOURS |
|---|---|---|
| MPUK3142 | Malaysian CITRA | 2 |
| MPUK3132 | Basic Malay for Foreigner | 2 |
| MPUK3312 | Nationhood Studies | 2 |
| MPUK3212 | Principles of Entrepreneurship | 2 |
| MPUK3152 | Integrity And Anti- Corruption | 2 |
| VXXXXXXX | Co-Curriculum | 2 |
| | **TOTAL** | **12** |

- Malaysian students are **REQUIRED** to take **MPUK3411 Bahasa Melayu Ilmiah** as a compulsory audit course and will not calculated in CGPA.
- All students must choose ONE (1) co-curricular area of interest, and fulfill the requirements of that chosen area.

## B. ENGLISH CORE COURSES (6 CREDIT HOURS)

Students are required to take the English Core Courses based on their MUET/IELTS/TOEFL results obtained during the admission application. Refer to **APPENDIX A.**

## C. PROGRAMME CORE COURSES (69 CREDIT HOURS)

| Code | Course | Credit Hours | Pre- requisite |
|---|---|---|---|
| BEEBK1013 | Principles of Economics | 3 | NONE |
| SQQMK1034 | Calculus I | 4 | NONE |
| SQQMK1063 | Discrete Mathematics | 3 | NONE |
| SQQMK1073 | Business Mathematics | 3 | SQQMK1034 Calculus I |
| SQQMK2023 | Linear Algebra | 3 | SQQMK1034 Calculus I **OR** SQQMK1023 Managerial Mathematics |
| SQQMK2034 | Calculus II | 4 | SQQMK1034 Calculus I |
| SQQMK2073 | Actuarial Mathematics I | 3 | SQQMK1073 Business Mathematics **AND** SQQSK1043 Probability and Statistics |
| SQQMK3024 | Mathematics Modelling | 4 | SQQMK3054 Numerical Analysis |
| SQQMK3034 | Multivariate Calculus | 4 | SQQMK2034 Calculus II |
| SQQMK3043 | Ordinary Differential Equation | 3 | SQQMK2034 Calculus II |
| SQQMK3054 | Numerical Analysis | 4 | SQQMK3043 Ordinary Differential Equation **AND** SQITK1013 Programming in Business Applications **or** STIAK1014 Introduction to Programming |
| SQQZK4994 | Academic Project | 4 | SQQMK3024 Mathematics Modelling **AND** SQQPK3023 Decision Modelling **AND** SQQSK2073 Regression Modelling |
| SQQMK3073 | Actuarial Mathematics II | 3 | SQQMK2073 Actuarial Mathematics I |
| SQQSK1033 | Data Exploratory and Generalisation | 3 | NONE |
| SQQSK1043 | Probability and Statistics | 3 | SQQMK1034 Calculus I |
| SQQSK2063 | Inferential Statistics | 3 | SQQSK1043 Probability and Statistics |
| SQQSK2073 | Regression Modelling | 3 | SQQSK2063 Inferential Statistics |
| SQQPK3023 | Decision Modelling | 3 | NONE |
| SQITK1013 | Programming in Business Applications | 3 | NONE |

| | | | |
|---|---|---|---|
| SQQXK4966 | Industrial Training | 6 | Have taken and passed all courses under the programme structure |

**D. BUSINESS PROGRAMME ELECTIVE (15 CREDIT HOURS)**

Students are required to choose **FIVE (5)** courses from the list below.

| Code | Course | Credit Hours | Pre- requisite |
|---|---|---|---|
| BPMMK1013 | Principles of Marketing | 3 | NONE |
| BPMMK3313 | Consumer Behavior and Analytic | 3 | BPMMK1013 Principles of Marketing |
| BPMMK2023 | Marketing Management | 3 | BPMMK1013 Principles of Marketing |
| BPMEK2013 | Entrepreneurial Behaviour and Skill | 3 | MPU1052 Principles of Entrepreneurship |
| BPMEK2023 | Creativity and Innovation | 3 | NONE |
| BPMEK3033 | E-Commerce | 3 | NONE |
| BPMEK2033 | Social Entrepreneurship | 3 | MPU1052 Principles of Entrepreneurship |
| BEEBK2013 | Microeconomics | 3 | BEEBK1013 Principles of Economics |
| BEEBK2023 | Macroeconomics | 3 | BEEBK1013 Principles of Economics |
| BWBBK1013 | Foundations of Banking | 3 | NONE |
| BWBBK2013 | Bank Management | 3 | NONE |
| BWRRK1013 | Risk and Insurance | 3 | NONE |
| BWRRK2043 | Insurance Company Operations | 3 | BWRRK1013 Risk and Insurance |
| BIMCK2013 | Fiqh Muamalat | 3 | NONE |
| BIMEK1013 | Theory and Practice of Islamic Business | 3 | NONE |
| BIBMK3043 | Islamic Banking Management | 3 | NONE |

**E. QUANTITATIVE PROGRAMME ELECTIVE (15 CREDIT HOURS)**

Students are required to choose **FIVE (5)** courses from the list below.

| Code | Course | Credit Hours | Pre- requisite |
|---|---|---|---|
| SQQPK3033 | Computer Modeling in Business | 3 | NONE |
| SQQPK3063 | System Dynamics Modeling | 3 | NONE |
| SQQPK3113 | Game Theory and Strategic Thinking | 3 | SQQMK1034 Calculus I |
| SQITK3083 | Data Visualization | 3 | NONE |
| SQITK3053 | Computer Aided Decision Making | 3 | SQITK1013 Programming in Business Applications |
| SQITK3063 | Data Mining | 3 | NONE |
| SQITK3043 | Data Modeling and Database | 3 | SQITK1013 Programming in Business Applications |
| SQQSK2083 | Sampling Techniques | 3 | SQQSK2063 Inferential Statistics |
| SQQSK3063 | Statistical Quality Improvement | 3 | SQQSK2063 Inferential Statistics |
| SQQSK3103 | Design and Analysis of Experiment | 3 | SQQSK2073 Regression Modelling |
| SQQSK3113 | Multivariate Analysis | 3 | SQQSK2073 Regression Modelling **AND** SQQMK2023 Linear Algebra |
| SQQSK3143 | Lifetime Data Analysis | 3 | SQQSK2073 Regression Modelling |
| SQQSK3153 | Non Parametric Methods | 3 | SQQSK2063 Inferential Statistics |
| SQQSK3163 | Forecasting Methods | 3 | SQQSK2073 Regression Modelling |
| SQQSK3173 | Official Statistics | 3 | SQQSK2063 Inferential Statistics |
| SQQSK3183 | Special Topics in Statistics | 3 | Depend on topics offered |

**F. FREE ELECTIVE (3 CREDIT HOURS)**

Students can choose ONE (1) course offered by OTHER SCHOOLS subject to the following conditions;

i. Must meet the prerequisite of the courses if such courses require prerequisites.
ii. **University Core, English Core** and **Foreign Language courses CANNOT** be considered as free elective courses.
iii. Equivalent courses **CANNOT** be considered as free elective courses. Refer **APPENDIX B.**

**PROGRAMME SUGGESTED COURSES BY SEMESTER FOR**
**BACHELOR OF SCIENCE WITH HONOURS (BUSINESS MATHEMATICS)**
**[B.Sc. (Hons) (Business Mathematics)]**

| Code | 1st Semester | Credit Hours | Code | 2nd Semester | Credit Hours |
|---|---|---|---|---|---|
| MPUK1042 | Nationhood Studies | 2 | MPUK3152 | Integrity and Anti- Corruption | 2 |
| BEEBK1013 | Principles of Economics | 3 | SQITK1013 | Programming in Business Applications | 3 |
| SQQMK1034 | Calculus I | 4 | SQQMK2034 | Calculus II | 4 |
| SQQMK1063 | Discrete Mathematics | 3 | SQQMK2023 | Linear Algebra | 3 |
| SQQSK1033 | Data Exploratory and Generalisation | 3 | SQQSK1043 | Probability and Statistics | 3 |
| MPBXXXX/ SBLEKXXXX | English Core | 3 | VXXXXXXX | Co-curriculum | 1 |
| VXXXXXXX | Co-curriculum | 1 | XXXXXXXXX | Business Programme Elective 1 | 3 |
| **TOTAL** | | **19** | **TOTAL** | | **19** |
| **Code** | **3rd Semester** | **Credit Hours** | **Code** | **4th Semester** | **Credit Hours** |
| MPUK3122 / **MPUK3132 | Malaysian Nation-State Aspiration/ **Basic Malay for Foreigner | 2 | SQQMK1073 | Business Mathematics | 3 |
| MPUK3212 | Principles of Entrepreneurship | 2 | SQQMK3054 | Numerical Analysis | 4 |
| SQQMK3034 | Multivariate Calculus | 4 | SQQSK2073 | Regression Modelling | 3 |
| SQQMK3043 | Ordinary Differential Equation | 3 | XXXXXXXXX | Business Programme Elective 2 | 3 |
| SQQSK2063 | Inferential Statistics | 3 | XXXXXXXXX | Business Programme Elective 3 | 3 |
| MPBXXXX/ SBLEKXXXX | English Core | 3 | XXXXXXXXX | Quantitative Programme Elective 2 | 3 |
| XXXXXXXXX | Quantitative Programme Elective 1 | 3 | MPUK3411 | Bahasa Melayu Ilmiah | 1 |
| **TOTAL** | | **20** | **TOTAL** | | **20** |
| **Code** | **5th Semester** | **Credit Hours** | **Code** | **6th Semester** | **Credit Hours** |
| MPUK3112 / **MPUK3142 | Falsafah dan Cabaran Semasa / **Malaysian CITRA | 2 | SQQMK3073 | Actuarial Mathematics II | 3 |
| SQQMK2073 | Actuarial Mathematics I | 3 | SQQZK4994 | Academic Project | 4 |
| SQQMK3024 | Mathematics Modelling | 4 | XXXXXXXXX | Business Programme Elective 5 | 3 |
| SQQPK3023 | Decision Modelling | 3 | XXXXXXXXX | Quantitative Programme Elective 4 | 3 |
| XXXXXXXXX | Business Programme Elective 4 | 3 | XXXXXXXXX | Quantitative Programme Elective 5 | 3 |
| XXXXXXXXX | Quantitative Programme Elective 3 | 3 | XXXXXXXXX | Free Elective | 3 |
| | | | | | |
| **TOTAL** | | **18** | **TOTAL** | | **19** |

| Code | 6th Semester (Short Semester) | Credit Hours |
|---|---|---|
| SQQXK4966 | Industrial Training | 6 |
| **TOTAL** | | **6** |

**Total Overall Credit Hours: 120 Credit Hours**

**Compulsory for International Student.

**PROGRAMME STRUCTURE FOR BACHELOR OF SCIENCE WITH HONOURS (INDUSTRIAL STATISTICS) [B.Sc.(Hons)(Industrial Statistics)]**

The Bachelor of Science with Honours (Industrial Statistics) consists of the following **FIVE (5)** components:

| COMPONENT | CREDIT HOURS |
|---|---|
| A. UNIVERSITY CORE COURSES | 12 |
| B. ENGLISH CORE COURSES | 6 |
| C. PROGRAMME CORE COURSES | 72 |
| D. PROGRAMME ELECTIVE | 30 |
| E. FREE ELECTIVE | 3 |
| **TOTAL CREDIT HOURS** | **123 HOURS** |

To be awarded Bachelor of Science with Honours (Industrial Statistics), a student is required to take and pass **a minimum of 123 credit hours** which include courses in the programme structure through the following structure:

**A. UNIVERSITY CORE COURSES (12 CREDIT HOURS)**

i. Malaysian Students

| CODE | COURSE | CREDIT HOURS |
|---|---|---|
| MPUK3112 | Falsafah dan Cabaran Semasa | 2 |
| MPUK3122 | Malaysian Nation-State Aspiration | 2 |
| MPUK3312 | Nationhood Studies | 2 |
| MPUK3212 | Principles of Entrepreneurship | 2 |
| MPUK3152 | Integrity And Anti- Corruption | 2 |
| MPUK3411 | Bahasa Melayu Ilmiah<br>**audit course* | 1 |
| VXXXXXXX | Co-Curriculum | 2 |
| | **TOTAL** | **13** |

ii. International Students

| CODE | COURSE | CREDIT HOURS |
|---|---|---|
| MPUK3142 | Malaysian CITRA | 2 |
| MPUK3132 | Basic Malay for Foreigner | 2 |
| MPUK3312 | Nationhood Studies | 2 |
| MPUK3212 | Principles of Entrepreneurship | 2 |
| MPUK3152 | Integrity And Anti- Corruption | 2 |
| VXXXXXXX | Co-Curriculum | 2 |
| | **TOTAL** | **12** |

- Malaysian students are **REQUIRED** to take **MPUK3411 Bahasa Melayu Ilmiah** as a compulsory audit course and will not calculated in CGPA.
- All students must choose ONE (1) co-curricular area of interest, and fulfill the requirements of that chosen area.

**B. ENGLISH CORE COURSES (6 CREDIT HOURS)**

Students are required to take the English Core Courses based on their MUET/IELTS/TOEFL results obtained during the admission application. Refer to **APPENDIX A.**

**C. PROGRAMME CORE COURSES (72 CREDIT HOURS)**

| Code | Course | Credit Hours | Pre- requisite |
|---|---|---|---|
| BEEBK1013 | PRINCIPLES OF ECONOMICS | 3 | NONE |
| BPMNK1013 | PRINCIPLES OF MANAGEMENT | 3 | NONE |
| SQQMK1034 | CALCULUS I | 4 | NONE |
| SQQMK2023 | LINEAR ALGEBRA | 3 | SQQMK1034 CALCULUS I OR SQQMK1023 MANAGERIAL MATHEMATICS |
| SQQMK2034 | CALCULUS II | 4 | SQQMK1034 CALCULUS I |
| SQQMK3043 | ORDINARY DIFFERENTIAL EQUATION | 3 | SQQMK2034 CALCULUS II |
| SQQPK3023 | DECISION MODELING | 3 | NONE |
| SQQSK1033 | DATA EXPLORATORY AND GENERALISATION | 3 | NONE |
| SQQSK1043 | PROBABILITY AND STATISTICS | 3 | SQQMK1034 CALCULUS I |
| SQQSK2063 | INFERENTIAL STATISTICS | 3 | SQQSK1043 PROBABILITY AND STATISTICS |
| SQQSK2073 | REGRESSION MODELLING | 3 | SQQSK2063 INFERENTIAL STATISTICS |
| SQQSK2083 | SAMPLING TECHNIQUES | 3 | SQQSK2063 INFERENTIAL STATISTICS |
| SQQSK3063 | STATISTICAL QUALITY IMPROVEMENT | 3 | SQQSK2063 INFERENTIAL STATISTICS |
| SQQSK3103 | DESIGN AND ANALYSIS OF EXPERIMENT | 3 | SQQSK2073 REGRESSION MODELLING |
| SQQSK3113 | MULTIVARIATE ANALYSIS | 3 | SQQSK2073 REGRESSION MODELLING AND SQQMK2023 LINEAR ALGEBRA |
| SQQXK49512 | INDUSTRIAL TRAINING | 12 | HAVE TAKEN ALL COURSES UNDER THE PROGRAMME STRUCTURE |
| SQQZK3992 | ACADEMIC PROJECT I | 2 | SQQSK2083 SAMPLING TECHNIQUES |
| SQQZK3994 | ACADEMIC PROJECT II | 4 | SQQZK3992 ACADEMIC PROJECT I |
| STIAK1014 | INTRODUCTION TO PROGRAMMING | 4 | NONE |
| STIDK1013 | INFORMATION SYSTEM IN ORGANIZATION | 3 | NONE |

## D. PROGRAMME ELECTIVES CORE (30 CREDIT HOURS)

### 1. INDUSTRIAL OPTIONS (18 CREDIT HOURS)

Students are **REQUIRED** to take **TWO** (**2**) courses for each industrial:

**a) Transportation Industry (6 Credit Hours)**

| Code | Course | Credit Hours | Pre- requisite |
|---|---|---|---|
| BJTLK2013 | FUNDAMENTAL OF TRANSPORT AND LOGISTICS | 3 | NONE |
| BJTLK3063 | RAIL TRANSPORT MANAGEMENT | 3 | BJTLK2013 FUNDAMENTAL OF TRANSPORT AND LOGISTICS |
| BJTLK3073 | SEAPORT MANAGEMENT | 3 | BJTLK2013 FUNDAMENTAL OF TRANSPORT AND LOGISTICS |
| BJTLK3083 | MARITIME MANAGEMENT | 3 | BJTLK2013 FUNDAMENTAL OF TRANSPORT AND LOGISTICS |

**AND**

**b) Manufacturing Industry (6 Credit Hours)**

| Code | Course | Credit Hours | Pre- requisite |
|---|---|---|---|
| BJTMK3043 | MANUFACTURING PLANNING AND CONTROL | 3 | BJTNK2033 PRODUCTION AND OPERATIONS MANAGEMENT |
| BJTMK3023 | INDUSTRIAL ENGINEERING | 3 | NONE |
| BJTMK3033 | MANUFACTURING SYSTEM ENGINEERING | 3 | BJTMK2023 MANUFACTURING TECHNOLOGY |
| BJTNK2033 | PRODUCTION AND OPERATIONS MANAGEMENT | 3 | SQQSK1033 DATA EXPLORATORY AND GENERALISATION |

**AND**

**c) Services Industry (6 Credit Hours)**

| Code | Course | Credit Hours | Pre- requisite |
|---|---|---|---|
| BPMHK2013 | HUMAN RESOURCE MANAGEMENT | 3 | BPMNK1013 PRINCIPLES OF MANAGEMENT |
| BPMHK3023 | STAFFING | 3 | BPMHK2013 HUMAN RESOURCE MANAGEMENT |
| BPMHK3063 | CHANGE MANAGEMENT | 3 | BPMHK2013 HUMAN RESOURCE MANAGEMENT |
| BPMHK3073 | NEGOTIATIONS | 3 | BPMHK2013 HUMAN RESOURCE MANAGEMENT |
| BPMHK3103 | INDUSTRIAL RELATIONS | 3 | BPMHK2013 HUMAN RESOURCE MANAGEMENT |

### 2. QUANTITATIVE OPTIONS (12 CREDIT HOURS)

Students are **REQUIRED** to choose **FOUR** (**4**) courses from the list below:

| Code | Course | Credit Hours | Pre- requisite |
|---|---|---|---|
| SQQMK3054 | NUMERICAL ANALYSIS | 4 | SQQMK3043 ORDINARY DIFFERENTIAL EQUATION **AND** STIAK1014 INTRODUCTION TO PROGRAMMING |
| SQQPK3033 | COMPUTER MODELING IN BUSINESS | 3 | NONE |
| SQQSK3143 | LIFETIME DATA ANALYSIS | 3 | SQQSK2073 REGRESSION MODELLING |
| SQQSK3153 | NON PARAMETRIC METHODS | 3 | SQQSK2063 INFERENTIAL STATISTICS |
| SQQSK3163 | FORECASTING METHODS | 3 | SQQSK2073 REGRESSION MODELLING |
| SQQSK3173 | OFFICIAL STATISTICS | 3 | SQQSK2063 INFERENTIAL STATISTICS |
| SQQSK3183 | SPECIAL TOPICS IN STATISTICS | 3 | DEPEND ON TOPICS OFFERED |

## E. FREE ELECTIVE (3 CREDIT HOURS)

Students can choose ONE (1) course offered by OTHER SCHOOLS subject to the following conditions;

iv. Must meet the prerequisite of the courses if such courses require prerequisites.
v. **University Core**, **English Core** and **Foreign Language courses CANNOT** be considered as free elective courses.
vi. Equivalent courses **CANNOT** be considered as free elective courses. Refer **APPENDIX B.**

**COURSE OFFERINGS SCHEDULE**

| Code | Course | SEM I | SEM II |
|---|---|---|---|
| SQQSK3143 | Lifetime Data Analysis | X | |
| SQQSK3153 | Non Parametric Methods | | X |
| SQQSK3173 | Official Statistics | X | |
| SQQSK3183 | Special Topics In Statistics | | X |

**PROGRAMME SUGGESTED COURSES BY SEMESTER FOR**
**BACHELOR OF SCIENCE WITH HONOURS (INDUSTRIAL STATISTICS)**
**[B.Sc. (Hons) (Industrial Statistics)]**

| Code | 1st Semester | Credit Hours | Code | 2nd Semester | Credit Hours |
|---|---|---|---|---|---|
| MPUK1042 | Nationhood Studies | 2 | MPUK3212 | Principles Of Entrepreneurship | 2 |
| BPMNK1013 | Principles Of Management | 3 | MPUK3122 / **MPUK3132 | Malaysian Nation-state Aspiration/ **Basic Malay For Foreigner | 2 |
| SQQMK1034 | Calculus I | 4 | MPBXXXX/ SBLEKXXXX | English Language Enrichment | 3 |
| SQQSK1033 | Data Exploratory And Generalisation | 3 | SQQMK2034 | Calculus II | 4 |
| STIDK1013 | Information System In Organization | 3 | SQQSK1043 | Probability And Statistics | 3 |
| MPBXXXX/ SBLEKXXXX | English Language Enrichment | 3 | BEEBK1013 | Principles Of Economics | 3 |
| VXXXXXXX | Co-curiculum | 1 | MPUK3152 | Integrity And Anti- Corruption | 2 |
| | | | VXXXXXXX | Co-curiculum | 1 |
| **TOTAL** | | **19** | **TOTAL** | | **20** |
| **Code** | **3rd Semester** | **Credit Hours** | **Code** | **4th Semester** | **Credit Hours** |
| SQQSK2063 | Inferential Statistics | 3 | SQQMK3043 | Ordinary Differential Equation | 3 |
| SQQMK2023 | Linear Algebra | 3 | SQQSK2073 | Regression Modelling | 3 |
| STIAK1014 | Introduction To Programming | 4 | SQQSK2083 | Sampling Techniques | 3 |
| MPUK3112 / **MPUK3142 | Falsafah Dan Cabaran Semasa / **Malaysian Citra | 2 | BJTXKXXX3 | Manufacturing Industry Course | 3 |
| BJTXKXXX3 | Manufacturing Industry Course | 3 | BPMHKXXX3 | Service Industry Course | 3 |
| BPMHK2013 | Human Resource Management | 3 | BJTLKXXX3 | Transportation Industry Course | 3 |
| MPUK3411 | Bahasa Melayu Ilmiah | 1 | | | |
| **TOTAL** | | **19** | **TOTAL** | | **18** |
| **Code** | **5th Semester** | **Credit Hours** | **Code** | **6th Semester** | **Credit Hours** |
| SQQSK3103 | Design And Analysis Of Experiment | 3 | SQQPK3023 | Decision Modeling | 3 |
| SQQSK3113 | Multivariate Analysis | 3 | SQQXXXXXX | Quantitative Options 3 | 3 |
| SQQSK3063 | Statistical Quality Improvement | 3 | SQQXXXXXX | Quantitative Options 4 | 3 |
| SQQXXXXXX | Quantitative Options 1 | 3 | SQQZK3994 | Academic Project II | 4 |
| SQQXXXXXX | Quantitative Options 2 | 3 | XXXXXXXX | Free Elective | 3 |
| BJTLKXXXX3 | Transportation Industry Course | 3 | | | |
| SQQZK3992 | Academic Project I | 2 | | | |
| **TOTAL** | | **20** | **TOTAL** | | **16** |
| Code | 7th Semester | Credit Hours | | | |
| SQQXK49512 | Industrial Training | 12 | | | |
| **TOTAL** | | **12** | | | |

**Total Overall Credit Hours: 123 Credit Hours**

**Compulsory for International Student.

## APPENDIX A

### ENGLISH CORE COURSES

A student is required to take English core courses according to these requirements:

| PATH (TOTAL CREDIT HOURS) AND COURSES SHOULD BE TAKEN | PRE REQUISITE | MUET | IELTS | TOEFL (iBT) | PTE ACADEMIC | CEFR |
|---|---|---|---|---|---|---|
| PATH 1 (9 CREDIT HOURS)<br>• MPB1013 BASIC ENGLISH PROFICIENCY (credit bearing)<br>• MPB2013 ENGLISH LANGUAGE ENRICHMENT I<br>• MPB3013 ENGLISH LANGUAGE ENRICHMENT II | Have taken English Intensive course for international students | BAND 1.0<br>BAND 2.5 | <4.0 | 0 - 30 | 0 - 29 | A2 |
| PATH 2 (6 CREDIT HOURS)<br>• MPB2013 ENGLISH LANGUAGE ENRICHMENT I<br>• MPB3013 ENGLISH LANGUAGE ENRICHMENT II | Have taken MPB1013 Basic English Proficiency | BAND 3.0<br>BAND 3.5 | 4.0 - 5.0 | 30 - 45 | 30 - 41 | B1 - B1+ |
| PATH 3 (6 CREDITS HOURS)<br>• MPB3013 ENGLISH LANGUAGE ENRICHMENT II<br>• **1 ESP COURSE** | Have taken MPB2013 English Language Enrichment I | BAND 4.5 | 6.0 – 6.5 | 60-93 | 50-63 | B2 |
| | | BAND 4.0 | 5.5 | 46 - 59 | 42 - 49 | B2+ |
| **LIST OF ESP COURSES**:<br>• SBLEK3023 English for Small Group Communication<br>• SBLEK3033 Report Writing<br>• SBLEK3043 Hospitality English<br>• SBLEK3053 Public Speaking Skills<br>• SBLEK3063 English for Professional Communication | Have taken MPB3013 English Language Enrichment II | | | | | |
| **EXEMPTED FROM ANY ENGLISH LANGUAGE COURSES**<br>*except for language courses that are listed as programme core courses | | BAND 5.0 | 7.0 - 8.0 | 94 - 114 | 64 - 82 | C1 |

## APPENDIX B

Students at School of Quantitative Sciences are prohibited from taking the following courses as free electives:

| No. | Code | Course |
|---|---|---|
| 1 | STIQK1013 | Mathematics for Information Technology |
| 2 | STTQK1013 | Mathematics for Computing |
| 3 | BEEQK1013 | Introduction to Mathematical Economics |
| 4 | BJTHK1103 | Mathematics for Technology |
| 5 | SQQMK1023 | Managerial Mathematics |
| 6 | STIQK1113 | Statistics for Information Technology |
| 7 | SSQLK1113 | Statistic for Social Science |
| 8 | BEEQK1023 | Introduction to Statistical Economics |
| 9 | BJTNK2013 | Statistical Technique in Decision Making |
| 10 | USQSK2013 | Managerial Statistics |
| 11 | SGDPK2013 | Statistics in Education |
| 12 | STTQK1113 | Discrete Mathematics |
| 13 | SQQSK1013 | Elementary Statistics |
| 14 | BWRRK3023 | Actuarial Science |
| 15 | SDCRK3083 | Interactive Data Visualization |